

Global Child Hazard Database 2025

Technical Documentation

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WASH, Climate and Environment Data Unit

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Abbreviations and Acronyms

ACLED - Armed Conflict Location & Event Data
ASI – Agricultural Stress Index
CCRI – Children’s Climate Risk Index
CCRI-DRM – Children’s Climate Risk Index – Disaster Risk Management
EPP – Emergency Preparedness Plan
ESA – European Space Agency
FAO – Food and Agriculture Organization
HDI – Human Development Index
IIASA – International Institute for Applied Systems Analysis
INFORM – Index for Risk Management
IPCC – Intergovernmental Panel on Climate Change
JRC – Joint Research Centre (European Commission)
MAP – Malaria Atlas Project
NASA – National Aeronautics and Space Administration
ND-GAIN – Notre Dame Global Adaptation Index
OECD – Organisation for Economic Co-operation and Development
P. falciparum – *Plasmodium falciparum* (malaria parasite species)
P. vivax – *Plasmodium vivax* (malaria parasite species)
SCAP – Sustainability and Climate Change Action Plan
SDG – Sustainable Development Goals
SDMX – Statistical Data and Metadata eXchange
SIDS – Small Island Developing States
SPEI – Standardized Precipitation Evapotranspiration Index
SPI – Standardized Precipitation Index
UNDRR – United Nations Office for Disaster Risk Reduction
UNICEF – United Nations Children’s Fund
UNSD – United Nations Statistics Division
VHI – Vegetation Health Index
VUB – Vrije Universiteit Brussel
WASH – Water, Sanitation and Hygiene
WHO – World Health Organization
WMO – World Meteorological Organization

UNICEF's first Global Child Hazard Database is designed to be a critical resource to provide comprehensive and reliable hazard data essential for safeguarding children's rights and well-being worldwide. It serves a multitude of vital purposes, enabling more effective and targeted interventions in the face of increasing global hazards.

Key applications of this database include:

- **Multiple Hazard Exposure Analysis:** The database facilitates detailed analysis of children's exposure to multiple hazards at varying intensities, helping stakeholders to pinpoint hotspot areas.
- **Children's Climate Risk Index (CCRI):** The database is foundational for the CCRI, providing the necessary hazard exposure data, allowing for a robust assessment of child-specific climate risks, especially at national level. For subnational assessments such as CCRI-DRM, the database could support hazard exposure data if country level data is unavailable.
- **Emergency Preparedness and Response:** By providing insights into potential hazard impacts and historical occurrences, the database is crucial for developing robust emergency preparedness plans, informing anticipatory action, and guiding immediate disaster response and recovery efforts.
- **Climate Financing:** The data can be leveraged to advocate for climate financing towards regions and communities most vulnerable to hazards, ensuring resources are allocated effectively to build resilience and implement adaptation strategies.
- **Programming and Targeting:** The database support the design and implementation of child-focused programs by identifying high-risk areas and populations, allowing for precise targeting of interventions to reduce children's vulnerability including providing estimates for the next UNICEF Strategic Plan. It can also be used for sub-national equity analysis and Climate Landscape Analysis.
- **Influencing International and National Policy:** By providing concrete, evidence-based data on child exposure to hazards, the database is an invaluable tool for advocating for and shaping policies at both national and international levels, driving legislative and strategic changes to protect children from various hazards.
- **Research and Academic Study:** The standardized and comprehensive nature of the database makes it a valuable resource for academic research, fostering a deeper understanding of the complex interactions between hazards, climate change, and child welfare.
- **Disaster Risk Reduction Strategies:** Understanding the spatial and temporal characteristics of hazards allows for the development and implementation of targeted disaster risk reduction strategies, focusing on prevention, mitigation, and adaptation measures.

This technical documentation serves as a live document, with new hazard layers and updates to be incorporated as data becomes available, ensuring the database remains a cutting-edge and adaptable tool in providing evidence.

While the first version of the database only includes climate hazards, other relevant hazards (geophysical, conflict) will be added in 2025 Q3 and potentially expanding to include environmental indicators in 2025 Q4 and beyond (as data becomes available).

Introduction

UNICEF plays a crucial role in protecting and promoting the rights of children worldwide. As children around the world are increasingly exposed to multiple hazards, access to reliable and comprehensive hazard data is essential for effective programming and advocacy. This handbook aims to provide an overview of the various types of hazard data available, as well as guidance on how to access, analyze, understand, visualize and utilize this data.

The primary audience for this handbook is all relevant UNICEF staff and partners. This is intended to be a live document and new hazard layers will be added/updated as data becomes available in the future.

Why do we need a standardized Global Child Hazard Database?

Standardization is needed to ensure a shared understanding of these diverse datasets with comparability and consistency and includes the following key areas.



Standardized Data Acquisition and Cataloging:

- A unified approach to cataloging all hazard data with clear decisions on authoritative and additional sources.
- Establish a global hazard data catalog including data sources and structured metadata.



Common Data Processing and Storage:

- Identify and understand the current data processing methods and improve clarity on how we store, process, analyze and disseminate data
- Detailed common data pipeline/architecture to ensure standard workflow



Enhanced Data Visualization:

- Effective data visualization techniques to communicate complex data insights to diverse audience; discuss different options for the visualization platform



Centralized documentation:

- Interactive, easy-to-access clear documentation of all data, methods and products – throughout the disaster and programme management cycle



Communication best practices:

- To improve clarity in terminologies used for both internal and external communications

Understanding Hazard Data

A hazard can be defined as any process, phenomenon or human activity that may cause loss of life, injury or other health impacts, property damage, social and economic disruption or environmental degradation.

Hazard Categories

Classification of hazards varies across different institutions and for the Global Child Hazard Database, UNDRR's terminology and classification schemes have been applied and are categorized into the following general categories:

- **Hydrometeorological hazards** are of atmospheric, hydrological or oceanographic origin. Examples are tropical cyclones (also known as typhoons and hurricanes); floods, including flash floods; drought; heatwaves and cold spells; and coastal storm surges. Hydrometeorological conditions may also be a factor in other hazards such as landslides, wildland fires, locust plagues, epidemics and in the transport and dispersal of toxic substances and volcanic eruption material (UNDRR, 2020).

It is important to note the following distinctions:

- o **Climate data** refers to measurements of atmospheric conditions over time and encompasses a wide range of meteorological variables – from temperature, precipitation, sea level rise and wind patterns. Climate data is used to better understand

long term climate patterns, trends, and future conditions. This could be both observed and/or modelled data and includes both historical data and future projections (more details under section 1.3).

- **Climate hazard** refers to natural events that are pre-dominantly hydro-meteorological in nature such as floods, tropical storms, heatwaves, wildfires and droughts. These hazards can be exacerbated by climate change, leading to increased frequency, intensity, and unpredictability. Understanding climate hazards is core to risk analysis, emergency preparedness and planning for effective mitigation and adaptation.
- **Climate change** refers to the change in climate over an extended period caused by natural processes such as volcanic eruptions or variations in solar radiation as well as human activities such as over burning fossil fuels, deforestation and over emission of greenhouse gases. There is adequate evidence that anthropogenic climate change is increasing the frequency and intensity of climate hazards as well as influence along with existing natural variability that causes climate hazards. In other words, not all climate hazards are results of human-induced climate change. Attribution to climate change is a complex process and at the time of this writing, there is still limited data available to differentiate the two.
- **Climate sensitive** hazards could be used to refer to air pollution and diseases (vector borne such as malaria, dengue, zika and water-borne such as cholera) which could be intensified because of climate crisis
- **Environmental Hazards** may include chemical, natural and biological hazards. They can be created by environmental degradation or physical or chemical pollution in the air, water and soil. However, many of the processes and phenomena that fall into this category may be termed drivers of hazard and risk rather than hazards in themselves, such as soil degradation, deforestation, loss of biodiversity, salinization and sea-level rise (UNDRR, 2020).
- **Geophysical hazards** originate from internal earth processes. Examples are earthquakes, volcanic activity and emissions, and related geophysical processes such as mass movements, landslides, rockslides, surface collapses and debris or mud flows. Hydrometeorological factors are important contributors to some of these processes. Tsunamis are difficult to categorize although they are triggered by undersea earthquakes and other geological events, they essentially become an oceanic process that is manifested as a coastal water-related hazard (UNDRR, 2020).

All the above are interconnected and contribute to improving our understanding of the triple planetary crisis. Each hazard often triggers a sub-set of hazards, for instance tropical cyclones (known as hurricanes in the Atlantic Ocean, cyclones in the Indian Ocean and typhoons in the Northern Pacific Ocean) can bring intense winds, storm surge and heavy rainfall, as well as trigger secondary hazards, for instance landslides. A series of triggering relationships can cause a domino or cascading effect (UNDRR, 2020).

Hazard Characteristics

Hazards occur at different magnitudes (intensity) over different time scales (duration) and can occur multiple times (frequency). In general, intensities can be expressed in terms of absolute measures such as km/hr for cyclone windspeed or m depth for flood. Return periods are used to combine both measures. In general, the longer the return period (the less frequent the hazard) the greater the intensity of the hazard (UNDRR, 2020). Interaction between multiple hazards could result in an overall impact that is greater than a single hazard happening at different times.

Know your hazard data type!

Once hazards are identified, the next step is to collect a range of hazard-related data that shows spatial and temporal characteristics. There are five different phases of hazard data to be considered depending on the use cases. While some can be substituted for others, not all can be used inter-changeably. The

process can be challenging due to data limitations and in determining how best to consider secondary hazards that can be triggered by a primary event.

1. Probabilistic/Modelled Data

Probabilistic hazard models are generated through computer simulations using complex mathematical models and rely on statistical analysis of past data. These models help the users to understand areas that could potentially be impacted by a specific hazard and are usually available in different Return Periods. A Return Period represents the average interval of time between occurrences of the event. It is important to note that this does not mean the event will happen exactly once every 100 years. Instead, it indicates a 1% probability of the event occurring in any given year. Similarly, a 10-year return period event has a 10% probability of occurring in any given year.

A 100-year return period has been chosen as the default as it has a high chance of an event of a considerable magnitude happening within a lifetime. For example, a 100-year flood is a flood that has a 1% probability of being equaled or exceeded in any given year. There is approximately a 63.4% chance of experiencing at least one 100-year flood within a 100-year period.

Return Period (years)	Probability of experiencing at least one event in a 100-year period
2	100%
5	100%
20	99.41%
50	86.74%
100	63.40%
500	18.14%
1000	9.52%

Used in risk assessments, preparedness, and planning.

2. Historical Observations

Historical observations provide a baseline for understanding the frequency, intensity, and spatial distribution of hazards. This data is usually collected over a significant period to document past occurrences, helping users to understand the spatial and temporal distribution of each hazard. For example, 20 years of observed flood data from the Ganges plains could help us understand the possible areas at risk for floods. It is to note that global historical observation databases for many hazards are largely incomplete and often not updated regularly with spatial extent.

Used in risk assessments, preparedness, and planning.

3. Forecast

Forecasts offer short-term predictions of specific climate hazard events, enabling early warning systems and helping communities prepare for potential impacts. Hurricane forecasts, wildfire risk assessments, and flood warnings are examples of climate forecasts.

Used in anticipatory action and early warning.

4. Impact

Impact data assesses the consequences of climate hazards on people, property, and the environment. Understanding impacts helps to assess vulnerability and develop effective mitigation and adaptation strategies. Economic losses, loss of life, damage to infrastructure, and ecosystem disruption are some of the potential impacts of climate hazards. For example, satellite-based flood observations are used for flood response immediately after an event.

Used in disaster response and recovery – also added to the historical observations.

5. Projected Data

Projected data are future predictions of climate hazards based on climate models and historical observations. Projections of sea-level rise, extreme precipitation events, and heatwave frequency are examples of projected data that help us anticipate future risks. While in the plans, data on climate projection is not included in the Global Child Hazard Database (as of writing).

Used in long-term programme planning and resilience (when reliable) – this is still largely in the research phase.

Hazard data type summary:

Hazard Data	Description	Use Cases
Probabilistic Model	Data generated by statistical models that estimate the likelihood of a specific hazard occurring.	Risk assessments such as Children Climate Risk Index (CCRI, CCRI-DRM) and Emergency Preparedness (EPP)
Historical Observation	Data collected from past events, such as weather stations, satellite imagery, and historical records.	Trend analysis, baseline establishment, vulnerability assessment
Forecast Data	Data predicting future conditions based on current observations and models.	Early warning systems, emergency response, short-term planning
Impact Data	Data quantifying the consequences of a hazard, such as economic losses, casualties, and environmental damage.	Damage assessment, resource allocation, emergency response
Projection Data	Data simulating future climate conditions under different scenarios of greenhouse gas emissions. There are limited global data availability (floods, storms, to name a few)	Long-term planning, adaptation strategies, climate change impact assessment

The first version of the Global Child Hazard Database includes probabilistic models and historical observations and does not include forecast, impact and projection data which could potentially be included in the subsequent versions depending on resource availability and needs.

What is an extreme event?

It is not straight forward to define extreme weather events. As a rule of thumb, if an event significantly deviates from the norm in terms of their intensity, duration and impact, it could be considered as extreme events and for each hazard, thresholds are determined using expert input.

What is climate attribution?

Attribution in climate science determines the human influence on specific weather events. It answers: "Did human-caused climate change make this event more likely or intense?" This involves comparing the event's probability and intensity in the current climate versus a hypothetical one without human greenhouse gas emissions.

Data Acquisition and Cataloging

Multiple data sources were considered for each hazard and were carefully evaluated. The final selection is identified based on the following pre-established criteria: All input data must be open, globally available for all countries, quantifiable and comparable but most importantly spatially distributed. The diagram below shows the six criteria used in data selection.



Proxy indicators are considered where global open data is not available. Quality dimension of basic data is considered to maximize the overall quality of the final CCRI, and within the constraints of data availability and include their relevance, accuracy, timeliness, accessibility, interpretability and coherence (OECD, 2008).

Hazard Data Processing

The functionality of this database is contingent upon precise hazard data processing. This section details the methodologies for data acquisition, standardization, quality control, and integration of diverse hazard datasets. These processes ensure data consistency, accuracy, and interoperability, enabling comprehensive analysis and supporting evidence-based decision-making.

In order to enable easier data processing for the whole world, we adopt a simple boundary simplification. If a country is smaller than Singapore, we do not simplify, if it is larger than Italy, we use 100m and if it larger than Argentina, we use 10km.

Children Population data:

WorldPop's global gridded children population estimate (Under 18) for 2024 at 100m spatial resolution was used for all exposure analysis in Global Child Hazard Database. The constrained population data product was used. Individual country population data is first downloaded and mosaicked into a single, global composite layer for further processing. Worldpop is chosen for its higher resolution, global availability and as it has the least bias in estimating rural population (Ritter, 2025).

Limitations:

WorldPop is a modeled product that disaggregates coarser census or projected data onto finer grids using various geospatial covariates; thus, the accuracy of its output is intrinsically linked to the age and reliability of its input data. While some WorldPop datasets are adjusted to align with UN WPP national totals, the underlying spatial distribution might still differ from official national census figures or the detailed demographic structures assumed by the UN's own projections, as WorldPop's modeling cannot fully replicate the granular information gathered through comprehensive national censuses.

Hazard data:

Flood

Floods are the natural hazard with the highest frequency and widest geographic distribution (UNISDR, 2017) and are described in terms of frequency and intensity (water extent and depth). There are three common types of flooding.

- Fluvial (riverine)
- Coastal (storm surge / sea level rise)
- Pluvial (surface water / flash floods)

Relevance:

Floods are a critical concern in climate change discussions due to their widespread impact on human and natural systems. According to the IPCC 2022 AR 6 report, food risks and societal damages are projected to increase with every increment of global warming (medium confidence) (IPCC, 2022). They exacerbate existing vulnerabilities, particularly in communities already facing socioeconomic challenges. Children are especially vulnerable to the impacts of floods. Floods can disrupt their access to clean water, sanitation, and healthcare, leading to increased risks of diseases. Additionally, floods can cause significant psychological stress and trauma, affecting children's mental health and development (IPCC, 2022).

Fluvial Flood

A fluvial or riverine flood is a rise, usually brief, in the water level of a stream or water body to a peak from which the water level recedes at a slower rate (WMO, 2012). The primary cause of a fluvial flood is an extended precipitation event that occurs at, or upstream from, the affected area and can also occur when traditional flood-control structures, such as levees and dikes, are overtopped.

Data source:

Fluvial flood has been extensively studied, and multiple global models are freely available at different return periods. JRC's gridded Global Flood Hazard model (Baugh, 2024) at 100-year return period and 90m spatial resolution has been chosen and water depth above 1cm (defined as data pixel value greater than or equal to in our processing pipeline) has been considered for exposure analysis.

Data processing:

Children's exposure to riverine flood hazard was calculated by simple overlay of the flood hazard and the high-resolution gridded global child population layer. The results were then summarized to different administrative boundaries as absolute and relative values for all the Member States and territories using zonal statistics.

Coverage: Global

Indicators:

- Absolute number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to riverine floods
- Relative number of children (total, girls, boys), people (total, women, men), normalized scores of exposures riverine floods

Data limitations: While major riverine floods have been modelled at higher resolution, data might not cover smaller rivers.

Coastal Flood

Coastal flooding is most frequently the result of storm surges and high winds coinciding with high tides (WMO, 2011).

Data source:

A few open, global coastal flood models were compared at different return periods. For Global Child Hazard Database, JRC's gridded Global Coastal Flood Hazard model at 100-year return period and 90m spatial resolution has been chosen. This is a binary layer, and all values identified as 1 are considered for exposure analysis.

Data processing:

Children's exposure to coastal flood hazard was calculated by simple overlay of the flood hazard and the high-resolution gridded global child population layer. The results were then summarized to different administrative boundaries as absolute and relative values for all the Member States and territories using zonal statistics. Coastal flood does not have any thresholding, it is a binary 0, 1 non-presence or presence of flood used to calculate exposure.

Coverage: Global

Indicators:

- Absolute number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to coastal floods
- Relative number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to coastal floods

Data limitations:

JRC's coastal flood hazard layer does not include intensity (depth) information, which could be crucial to better estimate the exposure to different impacts on children.

Drought

Drought is a prolonged dry period in the natural climate cycle that can occur anywhere in the world. It is typically a slow on-set phenomenon caused by a lack of rainfall and can occur in four stages that are usually categorized as four drought types.

- Meteorological drought
- Agricultural drought
- Hydrological drought
- Socio-economic drought

Relevance:

Drought has always been a part of the natural variability of Earth's climate. Climate change increasingly puts pressure on food production and access, especially in vulnerable regions, and undermines food security and nutrition (WMO, 2012). The increase in the frequency, intensity, and duration of droughts will increase risks to food security (IPCC, 2022). Children are especially impacted by various drought risks. The lack of water can compromise their access to clean drinking water and sanitation, increasing the risk of diseases. Additionally, droughts can lead to food shortages, impacting children's nutrition and overall health. The psychological stress and trauma associated with drought conditions can also affect children's mental health and development (IPCC, 2022).

Agricultural Drought

Agricultural drought occurs when there is insufficient soil moisture to meet the needs of a particular crop at a particular time (FAO, 2025).

Data source:

The Agricultural Stress Index (ASI) facilitates the early identification of cropped land with a high likelihood of water stress (drought). The Index is based on the integration of the Vegetation Health Index (VHI) in two dimensions that are critical in the assessment of a drought event in agriculture: temporal and spatial. The first step is a temporal averaging of the VHI, assessing the intensity and duration of dry periods occurring during the crop cycle at the pixel level. The second step determines the spatial extent of drought events by calculating the percentage of pixels in arable areas with a VHI value below 35 percent (this value was identified as a critical threshold in assessing the extent of drought in previous research by Kogan, 1995). Each administrative area is classified according to the percentage of the affected area to facilitate the quick interpretation of results (FAO, 2025). A 100-year return period was generated using the long-term average from 1984 - 2023 as a proxy as it depicts the percentage of arable land, within an administrative area, that has been affected by drought conditions over the entire cropping season.

Data processing:

Children's exposure to agricultural drought was calculated by simple overlay of the areas where the 100-year ASI return period with over 30% cropland is affected and the high-resolution gridded global

child population layer. The results were then summarized to different administrative boundaries as absolute and relative values for all the Member States and territories using zonal statistics.

Coverage: Global

Indicators:

- Absolute number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to agricultural droughts
- Relative number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to agricultural droughts

Data limitations:

Drought is one of the most complex natural hazards and the impact is felt in places that are in proximity as well as those areas that rely on the food supply. In other words, drought can occur in one place and the impact can be felt anywhere that depends on the crop production, even in adjacent or far away countries. For the lack of better model, the drought exposure estimates in the database assumes children living in areas close to the actual location of agricultural drought to be directly affected and does not consider children living in areas that depend on the crop production as their primary food source which is hard to model globally.

Meteorological Drought

Definition:

Meteorological drought occurs when dry weather patterns dominate an area, defined by the degree of dryness and the duration of the dry period (WMO, 2021).

Data Source:

The Standardized Precipitation Index (SPI) and the Standardized Precipitation and Evapotranspiration Index (SPEI) were sourced from the European Copernicus Global Drought Observatory, available at ~25 km spatial resolution.

- SPI measures the deviation of precipitation from the climatological average over a given accumulation period and is widely used to detect and characterize meteorological droughts.
- SPEI adds the effect of evapotranspiration and is better suited to assess droughts under climate change scenarios where temperature changes are important. Longer accumulation periods (e.g., 12 months) are associated with persistent droughts and may also reflect agricultural and hydrological drought conditions (Copernicus, 2025).

Data Processing:

- Monthly aggregated SPI and SPEI data using a 12-month accumulation period were used for the years 2014–2024.
- Areas experiencing meteorological drought were identified as those with SPI or SPEI < -1, corresponding to approximately the driest 10% of conditions.
- These areas were overlaid with a high-resolution gridded global child population dataset to estimate exposure.
- Zonal statistics were used to summarize both absolute counts and relative proportions of exposed populations (children and total population, disaggregated by sex) across multiple administrative boundaries.

Coverage: Global

Indicators:

- Absolute exposure of children to Meteorological hazards (counts):
 - Children (total, girls, boys) People (total, women, men)
 - Normalized exposure scores (SPI and SPEI)

- Relative exposure of children to Meteorological hazards (percent of population):
 - Children (total, girls, boys)
 - People (total, women, men)
 - Normalized exposure scores (SPI and SPEI)

Data Limitations:

- The drought datasets have coarser spatial resolution (~25 km) compared to some other hazard data layers such as ASI (Agricultural Stress Index).
- Return period analysis (e.g., Gumbel fitting for extreme dryness) was not satisfactory due to:
 - The sparse frequency of extreme dry events in the observation period
 - The distribution of SPI and SPEI values across both negative and positive ranges, complicating traditional extreme value fitting.
- Handling of Unresolved Geographies: To ensure the inclusion of Small Island Developing States (SIDS) and coastal territories in the multi-hazard composite index, a value of 0 was assigned to drought exposure (SPI/SPEI) where the global land-mask failed to resolve the landmass. These values should be interpreted as a lower-bound estimate necessitated by model resolution constraints. While these areas are reported as having 0% exposure in the global dataset, local climate conditions may vary. This reporting convention was chosen to allow these countries to be represented in the final CCRI ranking, as an NA value would have excluded them from the multi-hazard geometric mean calculation.

Tropical Storms

A tropical storm is a type of storm system characterized by a low-pressure center, a closed low-level atmospheric circulation, strong winds, and a spiral arrangement of thunderstorms that produce heavy rain (WMO, 2021). Tropical storm classification is adopted from WMO's guidance where wind speed greater than 63 km/hr is considered as a named tropical storm (WMO, 2023)

Relevance:

Tropical storms can cause significant damage through high winds, heavy rainfall, and storm surges, leading to flooding, property destruction, and loss of life. Climate change is increasing the intensity and frequency of tropical storms. Warmer sea surface temperatures and higher atmospheric moisture levels contribute to more powerful storms with greater precipitation. This intensification is linked to human-induced greenhouse gas emissions, which have led to observable changes in storm patterns and behaviors (IPCC, 2022). Children are particularly vulnerable to the effects of tropical storms. These events can disrupt their access to clean water, food, and healthcare, increasing the risk of malnutrition and diseases and can lead to psychological trauma.

Data source:

The tropical cyclone wind data is derived from UNEP GIRI and indicates the mean value of wind velocity for a return period of 100 years. The map has global coverage and a resolution of 6' (~11.1 kilometers at the equator) (CDRI, 2023). The tracks of historical tropical cyclones used in GIRI were obtained from the IBTrACS database and hazard analysis was divided by oceanic basis with different cut off years, based on data availability (GIRI, 2023)

Data processing:

Children's exposure to tropical storms was calculated by simple overlay of the areas where the wind speed is over 63 km/hr and the high-resolution gridded global child population layer. The results were then summarized to different administrative boundaries as absolute and relative values for all the Member States and territories using zonal statistics.

Coverage: Global

Indicators:

- Absolute number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to tropical storms
- Relative number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to tropical storms

Data limitations:

The accuracy of the probabilistic hazard model depends on the completeness of historical tropical cyclone data which varies from one ocean to another. Incomplete data can lead to underestimation or overestimation of storm exposure.

Heatwave and Extreme Heat

A heatwave can be defined as a period where local excess heat accumulates over a sequence of unusually hot days and nights (WMO, 2022). Heatwaves are complex phenomena and at present, there is no universal consensus on a global metric. The specific criteria for what constitute a heatwave can vary by region, but it generally involves temperatures significantly higher than the average for that area and time of year.

A heatwave is defined as any period of 3 days or more when the maximum temperature each day is in the top 10% of the local 15-day average and the following three dimensions were considered:

- Heatwave frequency refers to the number of heatwaves per year.
- Heatwave duration refers to the total number of days an event lasts.
- Heatwave severity refers to the temperature above the local 15-day average during the heatwave, expressed in degrees Celsius.

Extremely high temperatures (extremely hot days) are estimated when a day exceeds 35 degrees Celsius.

Relevance:

Temperature is one of the core indicators to describe climate and there is clear evidence that climate change is increasing global temperatures and causing historic heat waves all around the world, with higher frequencies, duration and severity (IPCC, 2022). Children are more vulnerable to the short- and long-term effects, especially from the heat stress caused by exposure to heat waves that can negatively affect health and well-being, especially for infants and young children. Children sweat less per kilogram than adults and have a higher metabolism, which means they get hotter quicker. School closures because of heatwaves are becoming increasingly common, leading to negative impact on children's education.

It is important to note that heatwaves themselves don't lead to negative health consequences in all situations. For instance, with the above definition, heatwave severity increasing in the Nordic countries could lead to ecological impact which in turn could affect children's ability to have a quality life. It could lead to increased probability of wildfires which could decrease the air quality.

Heat stress represents a significant and growing hazard, particularly for vulnerable populations such as children, due to its direct physiological impact and exacerbation of existing health conditions.¹ While the availability of global data on heat stress exposure is steadily increasing, facilitating broader analysis, a universally accepted global threshold for children's safe exposure levels currently remains undefined, highlighting a critical area for further research and policy development.

Data source:

Temperature input is calculated using daily aggregate temperature data from the ERA5 reanalysis (Muñoz, 2019). ERA5, produced by the Copernicus Climate Change Service (C3S) at ECMWF, is the fifth-generation atmospheric reanalysis of the global climate covering the period from January 1940 to present (ECMWF, 2025). A 100-year return period was generated from the 1960-2024 ERA5 LAND data.

To assess the extremity of climate-related heatwave indicators—including annual frequency, duration, severity, and the number of days exceeding 35°C—we estimated multi-year return levels using a non-parametric, empirical approach. For each pixel, annual values were sorted in descending order, and empirical return periods were assigned using Weibull plotting positions, i.e., $T=(n+1)/\text{rank}$, where n is the number of years. Return levels for specific periods (e.g., 10-, 30-, 50-, and 100-year events) were then derived by interpolating the observed annual extremes against their associated empirical return periods. This method avoids distributional assumptions (e.g., Gumbel or GEV) and is particularly suited for short observational records or count-based variables such as event frequency. The approach aligns with recommended practices in climate extremes analysis, especially when parametric assumptions may not be valid or robust (WMO, 2009; Wilks, 2011; Coles, 2001).

For land areas, we used the ERA5-Land dataset, which offers a spatial resolution of approximately 10 km. However, since the ERA5-Land product excludes Small Island Developing States (SIDS) and many island regions due to its land-sea masking, we used the standard ERA5 dataset at a coarser resolution of approximately 24 km for those areas. Data for SIDS was obtained for the same time period as the land areas and processed using the same methodology to ensure consistency across the analysis.

Data processing:

Children's exposure to individual dimensions was calculated by simple overlay of the areas where the individual dimension of heatwave is greater than the long-term average and the high-resolution gridded global children population layer. The results were then summarized to different administrative boundaries as absolute and relative values for all the Member States and territories using zonal statistics.

Coverage: Global

Indicators:

- Absolute number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to heatwave frequency
- Relative number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to heatwave frequency
- Absolute number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to heatwave duration
- Relative number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to heatwave duration
- Absolute number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to heatwave severity
- Relative number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to heatwave severity
- Absolute number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to extreme temperature
- Relative number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to extreme temperature

Data limitations:

While ERA5-Land offer enhanced resolution (approx. 9 km), it does not cover Small Island Developing States (SIDS) and ERA5's global layer was used at 25km resolution. A significant challenge lies in the lack of a universal definition for "heatwave" or "extreme heat," leading to inconsistencies across different data sources and complicating cross-regional comparisons. The exclusion of humidity metrics can lead to an underestimation of actual physiological heat stress. Observational data sparsity, especially in remote or less developed regions, introduces uncertainties, and current datasets often do not adequately capture the localized amplification of temperatures due to the Urban Heat Island effect. Finally, a critical gap remains in the absence of a globally accepted physiological

threshold for children's safe exposure to heat, hindering precise impact assessments for this highly vulnerable demographic.

Sand and Dust Storms

Sand and dust storms (SDS) are caused by intense winds over areas of arid soil that pick up large amounts of ground material into the atmosphere. The most significant dust sources globally are concentrated in arid and semi-arid regions, particularly major deserts such as the Sahara in Africa, the Gobi in Asia, and the Arabian Desert in the Middle East. They originate from natural sources like deserts, dry lake beds, and coastal regions with loose sediment. Human activities exacerbate the problem through construction, agriculture, and poor land management practices that strip vegetation and expose soil to wind erosion. Climate change amplifies the occurrence of sand and dust storms by altering weather patterns and reducing vegetation cover. (WMO, 2021).

Relevance:

Over the past few decades, land degradation has contributed significantly to the increased number and size of anthropogenic SDS sources. Current trends in deforestation, agricultural expansion, and more frequent and severe droughts and heatwaves make countries more susceptible to SDS hazards (UNCCD 2021). Exposure to dust can lead to respiratory problems, allergies, and other health issues and the disruption of daily activities and schooling due to poor air quality can also affect children's education and overall wellbeing.

Data source:

UNCCD's Sand and Dust Storm susceptibility layer at 1km resolution is used. It employs global datasets for four indicators to estimate the extent of source potential and derive source intensity values: (i) soil texture (proportion of sand, silt, and clay), (ii) soil moisture (absolute minimum value), (iii) soil temperature (absolute maximum value), and (iv) land cover (bare land fraction). (UNCCD, 2024)

Data processing:

Children's exposure to sand and dust storms was calculated by simple overlay of the areas where there is SDS susceptibility and the high-resolution gridded global child population layer. The results were then summarized to different administrative boundaries as absolute and relative values for all the Member States and territories using zonal statistics.

Coverage: Global

Indicator:

- Absolute number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to Sand and Dust Storms
- Relative number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to Sand and Dust Storms

Data limitations:

SDS susceptibility layer includes areas of original source but not where the wind could transport the particles. This could underestimate the overall global exposure. The adopted methods in SDS base map also have limited precision in identifying lower intensity sources due to the lack of data as well as the high degree of uncertainty associated with soil-related data (UNCCD, 2024).

Fires

Wildfires are uncontrolled burns of vegetation, including forests, shrublands, grasslands, savannas, and croplands. They can be caused by human activity or natural causes (UNDRR, 2020). Fires caused deliberately for land clearance (agriculture and ranching) or accidentally (lightning strikes and human error) are a major factor in land-cover variability and change and hence affect fluxes of energy and water to the atmosphere (WMO, 2025).

Relevance:

Fires can lead to significant loss of biodiversity, destruction of property, and air pollution but they can also contribute to climate change by releasing large amounts of carbon dioxide and other greenhouse gases into the atmosphere (IPCC, 2022). Climate change is increasing the frequency and intensity of fires. Higher temperatures, prolonged droughts, and changes in vegetation patterns create conditions that are more conducive to wildfires, and human activities such as land use changes and deforestation also exacerbate the risk of fires (IPCC, 2022). Children are particularly vulnerable as they are exposed to smoke and air pollutants from fires that can lead to respiratory problems, allergies, and other health issues. Fires also impact their overall well-being, education and could cause significant psychological trauma.

Data source:

Two indicators are used to better understand fires – Frequency and Intensity.

Fire frequency is obtained from NASA's Fire Information for Resource Management System (FIRMS) is used as a proxy for fire hazard. Average fire frequency from 2001-2023 has been considered. The MODIS Fire and Thermal Anomalies product is available from the Terra (MOD14) and Aqua (MYD14) satellites as well as a combined Terra and Aqua (MCD14). The MODIS Fire and Thermal Anomalies products are derived from the MODIS sensors onboard the Terra (MOD14) and Aqua (MYD14) satellites. A combined product, MCD14, is also available and merges detections from both platforms. The nominal spatial resolution of the MODIS fire detection is 1 km, and the temporal resolution is daily, with up to four observations per day globally (two from each satellite). The sensor resolution is 1 km, and the temporal resolution is daily.

Fire Radiative Power (FRP) is a measure of the energy released by a fire in the form of radiation. It is typically expressed in watts per square meter per steradian per micrometer ($W/m^2/sr/\mu m$). FRP is an important parameter for understanding the intensity and behavior of fires, as well as their impact on the environment. The historical fires are based on vegetation fires only to capture all climate related events, but it is possible that the data includes human-induced vegetation fires.

Data processing:

Children's exposure to fires was calculated by simple overlay of the areas where there is fire frequency above 50 and fire intensity above 10 and the high-resolution gridded global child population layer. The results were then summarized to different administrative boundaries as absolute and relative values for all the Member States and territories using zonal statistics.

Coverage: Global

Indicator:

- Absolute number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to fire frequency
- Relative number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to fires frequency
- Absolute number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to fire intensity
- Relative number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to fires intensity

Data limitations:

Fire frequency and intensity are limited in identifying all areas that have potential to experience fires. It is adopted as a proxy for the lack of high-resolution global hazard data for Fire Weather Index (FWI) – a combination of which could give a better picture of fire hazard.

Air Pollution

Air pollution refers to the presence of substances in the atmosphere that are harmful to human health and the environment. These substances can include particulate matter, nitrogen dioxide, sulfur dioxide, carbon monoxide, and ozone (WHO, 2021).

Relevance:

Climate change and air pollution are closely linked. Many activities that produce greenhouse gases also emit air pollutants, which can act as short-lived climate forcers and can warm or cool the Earth's climate over shorter time scales (IPCC, 2022). Air pollution threatens children's health and is the biggest environmental health risk factor (UNICEF, 2024). Exposure to pollutants can lead to respiratory problems, asthma, and other health issues. Poor air quality can also affect children's development and cognitive function. Air pollution was the second leading risk factor for death among children under five in 2021, after malnutrition.

Particulate Matter 2.5 (PM2.5):

Fine particle air pollution, or PM_{2.5}, refers to airborne particles measuring less than 2.5 micrometers in diameter that could be emitted from vehicles, residential fuel use, coal-burning power plants, agricultural and industrial activities, waste burning, wildfires, and many other human and natural sources. Among the key air pollutants that are currently measured, long-term exposure to PM_{2.5} is the most consistent and accurate predictor of poor health outcomes across populations. (Health Effects Institute, 2024)

Data source:

Data from the Atmospheric Composition Analysis Group (ACAG) were global concentrations of PM_{2.5} estimated using information from satellite-, simulation- and monitor-based sources was used (Shen et al, 2024). Following the established WHO standards for air quality guidance, a threshold of 5 µg/m³ is adopted (WHO, 2021) and to account for short term variations such as decrease in air pollution during covid-19 pandemic, a ten-year average was considered following expert advice.

Data processing:

Children's exposure to particulate matter 2.5 was calculated by simple overlay of the areas where there is observed values of a ten-year average (2012 – 2022) was greater than 5 µg/m³ and the high-resolution gridded global child population layer. The results were then summarized to different administrative boundaries as absolute and relative values for all the Member States and territories using zonal statistics.

Coverage: Global

Indicator:

- Absolute number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to air pollution
- Relative number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to air pollution

Data limitations:

On-the-ground air quality monitoring stations are few and far between in many regions of the world, particularly in low- and middle-income countries which could affect the results and hence the exposure analysis. As noted above, there are several other air pollutants that are not included due to a lack of reliably modelled long-term global data at the time of writing. ACAG data is currently unavailable in higher latitudes, so not fully global at the moment and could be an underrepresentation for some countries.

Vector borne diseases

Vector borne disease are human illnesses caused by parasites, viruses, and bacteria that are transmitted by vectors.

Relevance:

Climate change can alter the distribution and abundance of vectors, leading to changes in the incidence and geographic range of these diseases. Higher temperatures combined with land use/land cover change are making more areas suitable for the transmission of vector-borne diseases (IPCC, 2022). Children are particularly susceptible to severe health problems from vector-borne diseases. For instance, malaria can cause high fever, anemia, and neurological complications, which are especially dangerous for young children and can also lead to increased mortality (WHO, 2024). Illnesses can also lead to malnutrition, disrupt education and affect a child's overall well-being.

Note: While there is various vector borne diseases such as dengue, zika and others, due to lack of global data availability, only Malaria was considered for the global child hazard database.

Malaria

Malaria is a life-threatening disease caused by parasites transmitted to humans through the bites of infected female Anopheles mosquitoes (WHO, 2021). There are 5 *Plasmodium* parasite species that cause malaria in humans and 2 of these species – *P. falciparum* and *P. vivax* – pose the greatest threat. *P. falciparum* is the deadliest malaria parasite and the most prevalent on the African continent. *P. vivax* is the dominant malaria parasite in most countries outside of sub-Saharan Africa (WHO, 2021).

Data source:

Gridded long-term estimates were downloaded from Malaria Atlas Project (MAP, 2025) and estimated from the global malariometric data (Pfeffer et al, 2018)

- Gridded average number of newly diagnosed *Plasmodium falciparum* cases per 1,000 population from 2012-2022 is used as a proxy for the spread of Malaria by *Plasmodium falciparum*
- Gridded average number of newly diagnosed *Plasmodium vivax* cases per 1,000 population from 2012-2022 is used as a proxy for the spread of Malaria by *Plasmodium vivax*

Data processing:

All children living in areas with more than 1 case is estimated as children exposed to Malaria (individually calculated for each) and overlaid with high-resolution gridded global children population layer. The results were then summarized to different administrative boundaries as absolute and relative values for all the Member States and territories using zonal statistics.

Coverage: Global

Indicator:

- Absolute number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to Malaria (*Plasmodium Falciparum*)
- Relative number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to Malaria (*Plasmodium Falciparum*)
- Absolute number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to Malaria (*Plasmodium Vivax*)
- Relative number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to Malaria (*Plasmodium Vivax*)

Data limitations:

Due to limitations in data availability, Malaria indicators only include estimates based on reported cases and does not include areas susceptible to Malaria based on climate indicators. It is highly recommended to use a combination of both long-term cases and susceptibility.

Earthquake

Data source: The Global Earthquake Model (GEM) Global Seismic Hazard Map (version 2023.1). K. Johnson, M. Villani, K. Bayliss, C. Brooks, S. Chandrasekhar, T. Chartier, Y. Chen, J. Garcia-Pelaez, R. Gee, R. Styron, A. Rood, M. Simionato, M. Pagani (2023). Global Earthquake Model (GEM) Seismic Hazard Map (version 2023.1 - June 2023), DOI: <https://doi.org/10.5281/zenodo.8409647>

Data processing:

The Global Earthquake Model (GEM) Global Seismic Hazard Map (version 2023.1) represents the spatial distribution of peak ground acceleration (PGA) expected to be exceeded with a 10 percent probability in 50 years, which is equivalent to a 475 year return period, for reference rock conditions with a V_{s30} value of 760 to 800 metres per second. To produce this layer, GEM scientists compiled and harmonized national and regional probabilistic seismic hazard models developed by multiple institutions and projects around the world. These models were integrated into a consistent global framework to create a unified hazard surface. Version 2023.1 updates the 2018 release and incorporates substantial model improvements across most regions, along with an increase in spatial resolution of approximately two and a half times, resulting in a more detailed and current representation of global seismic hazard on rock.

Two shaking intensity thresholds were used to estimate the number of children exposed to earthquakes, based on perceived shaking from the Mercalli scale and the associated potential for damage. Strong shaking, corresponding to intensity VI, is associated with a PGA of 0.09 g, while Severe shaking, corresponding to intensity VIII, is associated with a PGA of 0.34 g.

Coverage: Global

Indicator:

- Absolute number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to Earthquakes
- Relative number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to Earthquakes

Landslide

Data source: Worldbank/ Global Facility for Disaster Reduction and Recovery (GFDRR) rainfall triggered landslides (version 2)

Data processing: The rainfall-based Global Landslide Hazard Map is a worldwide, gridded dataset that estimates the annual frequency of significant landslides per square kilometer, based solely on rainfall-triggered events. Developed for the World Bank and GFDRR by Arup, the map combines a global landslide inventory from 1980 to 2018, NASA landslide-susceptibility data, and machine-learning modelling to produce a consistent, quantitative global assessment.

The dataset covers all land areas between 60°S and 72°N and focuses on “significant” landslides events large enough to be reported if they occurred in populated areas (generally >100 m²). It supports applications such as hazard screening, exposure assessment at national to global scales, and estimating expected annual impacts on people and infrastructure.

Two annual frequency thresholds of 0.01 and 0.001 landslides per square kilometer were applied to identify children exposed to rainfall-induced landslides. These thresholds correspond to the Medium to High and Moderate landslide hazard classes in the global landslide hazard dataset.

Coverage: Global

Indicator:

- Absolute number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to Landslides
- Relative number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to Landslides

Volcano

Data source: Smithsonian Institution's Global Volcanism Program (GVP) active volcanoes (1800 to April 2025)

Data processing: The volcano hazard layer was developed using the Smithsonian Institution's Global Volcanism Program (GVP) database, which identifies active volcanoes as those with at least one documented eruption since 1800. A 100-kilometre buffer was generated around each active volcano to represent the area of potential exposure. All buffered areas were then merged to create a single exposure surface, ensuring that overlapping zones did not result in double-counting. Children located within the combined 100-kilometre exposure zone were counted once, even if they were within a range of multiple volcanoes.

Coverage: Global

Indicator:

- Absolute number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to Volcanoes
- Relative number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to Volcanoes

Conflict

Data source: Armed Conflict Location & Event Data Project (ACLED) - Conflict events and fatalities (Sept 2020 to Aug 2025)

Data processing: The conflict hazard layer was developed using georeferenced point data from the Armed Conflict Location & Event Data Project (ACLED), capturing all recorded conflict events and associated fatalities between September 2020 and August 2025. Only ACLED events classified as Battles, Violence against civilians, or Explosions/Remote violence were included in the analysis. Event and fatality counts were collected for each location and processed to create standardized values that minimize the impact of exceptionally high or low numbers. These metrics were normalized to a scale from 0 to 1, and a geometric mean was calculated using the normalized event and fatality scores per point. Next, the resulting index was normalized to a range of 0 to 10 through a min-max method and mapped onto a global 0.1-degree grid. For each grid cell, the highest value was used, resulting in a consistent global spatial layer that represents conflict intensity.

To identify areas considered exposed to conflict, a conflict index threshold of 0.01 was applied. Grid cells with index values at or above this threshold were classified as exposed, and all children residing within these cells were counted as experiencing conflict exposure.

Coverage: Global

Indicator:

- Absolute number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to Conflict
- Relative number of children (total, girls, boys), people (total, women, men), normalized scores of exposures to Conflict

Pixel-level multi-hazard Analysis

This section outlines the process of calculating the multi-hazard exposure at the pixel level to assess the compounding exposure of child populations to multiple climate and climate-sensitive hazards. Pixel-level analysis is performed to capture fine-grained variations in climate risks within a country, enabling the identification of hotspots that national averages may overlook. This approach allows for a more precise understanding of climate risks at both local and national levels, with the flexibility to aggregate data to administrative levels 0, 1, or 2 based on the specific use case.

Multi-hazard count

Definition and Purpose

The Multi-hazard Count indicator quantifies the number of distinct climate and climate-sensitive hazards that are likely to occur in the same location and overlap with child populations. It is computed at high spatial resolution (up to 100 meters), allowing for fine-grained analysis of compounded climate risks.

Each hazard is processed with applied thresholds to identify areas where its intensity exceeds a meaningful level of risk. These binary hazard masks are then stacked to calculate, per pixel, the number of different hazards present. This results in a hazard count map, where pixel values reflect the number of hazards a given location is exposed to.

To make the indicator child-focused, this multi-hazard count layer is spatially filtered using high-resolution child population data. The result is a map that highlights only the areas where children are present and exposed to one or more hazards. While the pixel values indicate the number of overlapping hazards, they only appear in locations with non-zero child population.

Child-Focused Use Case

By filtering the hazard count by child population, the indicator serves as a tool to identify areas where children face compounded climate risks. It enables targeted planning for health, education, and protection interventions, and supports more equitable resource allocation.

Aggregation and Applications

The pixel-level results can be aggregated to administrative units (e.g., district, province) to calculate:

- Total number of children exposed to one or more hazards
- Number of children exposed to 1, 2, 3, 4 or more concurrent hazards
- Distribution of hazard count among children within a region

This indicator supports decision-making in disaster risk reduction, climate adaptation, and child-focused program design.

Multi-hazard intensity

Definition and Purpose

The Multi-Hazard Intensity indicator evaluates not only whether a location is exposed to hazards, but how severe or extreme those hazards are. This high-resolution metric—computed at approximately 10 km spatial resolution—integrates the magnitudes of various climate and climate-sensitive hazards into a composite risk score for each location. Unlike a simple count of hazard types, this indicator distinguishes between areas experiencing frequent but moderate events and those exposed to less frequent but highly destructive climate extremes.

By capturing intensity, this metric offers a more nuanced understanding of risk, enabling local governments, humanitarian actors, and climate resilience planners to better prioritize areas that face more severe threats—even if only from one or two dominant hazards.

Data Processing:

Each hazard layer—such as heatwave frequency, drought severity, or flood intensity—is first cleaned and normalized to ensure comparability across different hazard types. This process involves removing non-informative values (e.g., zeros, NaNs), resampling to a consistent 10 km grid using a maximum-value method (to preserve peaks), and applying log transformations followed by z-score scaling to standardize the distributions. Pre-scaled datasets like SPI or soil moisture anomaly are directly included after trimming outliers to improve distribution symmetry.

Once the data layers are harmonized, they are stacked into a multi-dimensional array, where each pixel holds values representing the magnitude of different hazards. Principal Component Analysis (PCA) is then used to reduce this complex dataset to its most informative components—capturing over 90% of total variance with just 11 principal components. These components provide a way to synthesize multiple hazard signals while preserving key differences between regions.

From the PCA results, a weighted combination of principal component loadings—informed by each component's explained variance—is used to derive a final composite hazard intensity score for each land pixel. This produces a raw indicator of multi-hazard intensity, which is then normalized to a 0–10 scale using min-max scaling. The final result is a single raster layer where each valid land pixel carries a score representing the cumulative severity of climate risks in that location.

Note: We started with multiple hazard indicators across different dimensions, which created high correlation and redundancy in the dataset. To address this, we applied **PCA and retained 6 principal components**, which together explain **86% of the overall variation** in the hazard dataset.

The purpose of PCA was twofold:

Dimension reduction – to minimize multicollinearity among highly correlated hazard variables and simplify the structure without losing most of the information.

Deriving data-driven weights – the loadings from the principal components provide a statistically consistent way to reflect the relative importance of each hazard indicator when constructing the overall hazard index.

Regarding the idea of simply adding standardized indicators: while an additive index is more interpretable, it assumes equal weights for all hazards, which may not reflect their actual variance and influence in the dataset. PCA helps us avoid arbitrary weighting and ensures that the combined index captures the dominant patterns in the data.

As for counting hazards of different intensities even if correlated: PCA doesn't discard them but instead combines their shared variance into components, so the final measure still accounts for intensity differences without overemphasizing correlated indicators.

Hazard Combination

Recent studies indicate that the co-occurrence of climate hazards is becoming more frequent. From 2003–2012 to 2012–2021, the number of individuals exposed to three or more hazards increased by 69%*. While many hazard combinations are still uncommon, their rising frequency shows the urgent need for a deeper understanding of these overlapping risks to effectively design and implement climate-resilient programs.

Child population exposure to different Hazards Combinations

The analysis quantifies how many children are exposed to multiple climate and environmental hazards and their combinations. We start by loading child population data and hazard layers (e.g., floods, droughts, heatwaves, air pollution), aligning them to a common projection and scale. For each hazard, we apply a threshold to identify high-risk areas and mask the child population within those

zones, creating hazard-specific exposure layers. These layers are then grouped into broader hazard topics and combined using a binary encoding system to generate a combination ID, which uniquely represents overlapping hazard exposures at each location. Finally, we aggregate these results at the country level, decoding the combinations to show which hazards overlap, how many hazards are involved (Single, Double, Triple, Multiple), and the total child population exposed.

Finally, we compute totals for all three-hazard combinations, both including and excluding air pollution and malaria, to highlight the difference between purely climate-related risks and combined climate-health risks. This approach provides a detailed view of compound risks and their impact on children.

Child Population Exposure Estimate

The thresholds are applied to categorize child populations based on different levels of climate risk exposure. The child population exposure is calculated at various risk thresholds, such as the mean and percentiles, to estimate the number of children affected at different risk levels.

Child-Focused Application

To center the metric on children, the composite intensity surface is overlaid with high-resolution child population data. This overlay allows for the estimation of the number of children exposed to varying levels of climate hazard severity—from average exposure to the top 1% of most at-risk zones.

This approach highlights not only where the most intense risks occur but also who is affected—making it a powerful tool for guiding investments in child protection, school infrastructure resilience, and health services. It also allows for threshold-based analysis, helping governments and agencies understand the scale of exposure at critical risk levels (e.g., above the 90th percentile).

Sensitivity and Limitations

The method allows for robust sensitivity testing, including the effect of resampling strategies, the number of PCA components used, and alternative normalization approaches. However, it also carries assumptions—such as the linearity of PCA and the weighting of components—which must be carefully validated in different contexts.

Despite these limitations, the Multi-Hazard Intensity metric offers a scalable, data-driven approach to understanding where high-severity climate risks intersect with vulnerable populations—a critical step toward effective and equitable climate resilience planning.

Data Processing:

The indicators presented in the table below were derived from gridded datasets and aggregated at the global administrative level 2 for all countries. These indicators were exported as both wide and long format spreadsheets and shapefiles, making the data accessible to users with varying technical expertise. The data was organized into three separate data flows—Exposure, Hazard, and Population—to facilitate straightforward uploads and retrievals via SDMX.

Indicator	Description	Indicator type	Unit
Hazard mean	Average intensity value of the hazard across the assessed administrative area. Higher numbers mean the physical danger is generally greater.	Hazard Intensity	Value
Hazard standard deviation	How much the intensity varies within the assessed administrative area. A large value means some locations are much riskier than others.	Hazard Intensity	Value

Hazard maximum	Highest hazard intensity within the administrative area. Higher values indicate severe localized impacts.	Hazard Intensity	Value
Hazard minimum	Lowest hazard intensity within the administrative area. Lower values represent minimal hazard intensity.	Hazard Intensity	Value
Hazard median	Middle value of hazard intensity when ranked from lowest to highest within the administrative area. Higher medians suggest widespread moderate–high intensity.	Hazard Intensity	Value
Hazard Severity Index (global)	A 0–10 index scaled using global minimum and maximum hazard intensity values. Higher values indicate more severe global hazard levels.	Hazard Index	Index (0–10)
Hazard Severity Index (country)	A 0–10 index scaled using country-specific minimum and maximum hazard intensity values. Higher values indicate severe national-level hazard conditions.	Hazard Index	Index (0–10)
Total population exposed to hazard	Estimated number of people residing in hazard-exposed areas.	Absolute Exposure	Persons
Total male population exposed to hazard	Estimated number of males residing in hazard-exposed areas.	Absolute Exposure	Persons
Total female population exposed to hazard	Estimated number of females residing in hazard-exposed areas.	Absolute Exposure	Persons
Total children (under 18) exposed to hazard	Estimated number of children (under 18 years) residing in hazard-exposed areas.	Absolute Exposure	Persons
Total male children (under 18) exposed to hazard	Estimated number of male children (under 18 years) residing in hazard-exposed areas.	Absolute Exposure	Persons
Total female children (under 18) exposed to hazard	Estimated number of female children (under 18 years) residing in hazard-exposed areas.	Absolute Exposure	Persons
Percentage of population exposed to hazard	Estimated proportion of the total population of the administrative area residing in hazard-exposed areas.	Relative Exposure	Percent (%)
Percentage of male population exposed to hazard	Estimated proportion of all males in the administrative area residing in hazard-exposed areas.	Relative Exposure	Percent (%)

Percentage of female population exposed to hazard	Estimated proportion of all females in the administrative area residing in hazard-exposed areas.	Relative Exposure	Percent (%)
Percentage of children (under 18) exposed to hazard	Estimated proportion of all children (under 18) in the administrative area residing in hazard-exposed areas.	Relative Exposure	Percent (%)
Percentage of male children (under 18) exposed to hazard	Estimated proportion of male children (under 18) in the administrative area residing in hazard-exposed areas.	Relative Exposure	Percent (%)
Percentage of male children (under 18) exposed to hazard	Estimated proportion of female children (under 18) in the administrative area residing in hazard-exposed areas.	Relative Exposure	Percent (%)
Global exposure index	A 0–10 index that combines log-scaled absolute exposure and percentage-based relative exposure (each robust min–max normalized) via the geometric mean, identifying locations where both the number and proportion of people exposed are simultaneously high.	Exposure Index	Index (0–10)
Country exposure index	A 0–10 index that combines log-scaled absolute exposure and percentage-based relative exposure (each robust min–max normalized) via the geometric mean, identifying locations within a country where both the number and proportion of people exposed are simultaneously high.	Exposure Index	Index (0–10)
Global exposure class	Categorization of global exposure index scores into five classes indicating relative levels of population exposure, from Very low to Very high.	Exposure Class	Class label
Country exposure class	Categorization of country-specific exposure index scores into five classes, clearly conveying relative exposure levels at the national scale.	Exposure Class	Class label

Annex

Data licensing and copyrights

All data used in CCRI 2025 is open source.

Data Providers

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Appendix

For more information on metadata, please refer to [Global Child Hazard Data](#) catalog.

References

- Baugh, Calum; Colonese, Juan; D'Angelo, Claudia; Dottori, Francesco; Neal, Jeffrey; Prudhomme, Christel; Salamon, Peter (2024): Global River Flood Hazard Maps. European Commission, Joint Research Centre (JRC) [Dataset] PID: http://data.europa.eu/89h/jrc-floods-floodmapgl_rp50y-tif. Accessed 17 Mar. 2025
- Bondarenko M., Priyatikanto R., Tejedor-Garavito N., Zhang W., McKeen T., Cunningham A., Woods T., Hilton J., Cihan D., Nosatiuk B., Brinkhoff T., Tatem A., Sorichetta A.. *Estimates of total number of people under the age of 18 years old and broken down by male and female for each year 2015-2030 at a resolution of 3 arc (approximately 100m at the equator) R2025A version v1. Global Demographic Data Project - Funded by The Bill and Melinda Gates Foundation (INV-045237). WorldPop - School of Geography and Environmental Science, University of Southampton*. DOI:10.5258/SOTON/WP00847
- CDRI (2023). Global Infrastructure Resilience: Capturing the resilience dividend - A Biennial Report from the Coalition for Disaster Resilient Infrastructure, New Delhi.
- Copernicus Global Drought Observation. "Standard Precipitation Index." EDO and GDO Indicator Factsheet. https://drought.emergency.copernicus.eu/data/factsheets/factsheet_spi.pdf. Accessed 17 Mar. 2025
- Coles, S. (2001). *An Introduction to Statistical Modeling of Extreme Values*. Springer.
- Dottori F; Salamon P; Alfieri L; Bianchi A; Feyen L; Hirpa F; Lorini V. Flood Hazard Maps at European and Global Scale. European Commission; 2016. JRC103765. <https://publications.jrc.ec.europa.eu/repository/handle/JRC103765>
- Dottori, F., Salamon, P., Bianchi, A., Alfieri, L., Hirpa, F.A., Feyen, L., 2016. Development and evaluation of a framework for global flood hazard mapping. *Adv. Water Resour.*94, 87–102. <https://doi.org/10.1016/j.advwatres.2016.05.002>.
- Health Effects Institute. 2024. State of Global Air 2024. Special Report. Boston, MA:Health Effects Institute. <https://www.stateofglobalair.org/resources/report/state-global-air-report-2024>. Accessed 17 Mar 2025
- IPCC, 2022: Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [H.-O. Pörtner, D.C. Roberts, M. Tignor, E.S. Poloczanska, K. Mintenbeck, A. Alegría, M. Craig, S. Langsdorf, S. Löschke, V. Möller, A. Okem, B. Rama (eds.)]. Cambridge University Press. Cambridge University Press, Cambridge, UK and New York, NY, USA, 3056 pp., doi:10.1017/9781009325844.
- Intergovernmental Panel on Climate Change. (2021). *Climate Change 2021: The Physical Science Basis*. Retrieved from <https://www.ipcc.ch/report/ar6/wg1/chapter/chapter-6/>
- Intergovernmental Panel on Climate Change. (2022). *Climate Change 2022: Impacts, Adaptation and Vulnerability*. Retrieved from <https://www.ipcc.ch/report/ar6/wg2/chapter/chapter-4/>. Accessed 17 Mar 2025.
- GIRI, 2023. Multi-hazard Disaster Risk Model of Infrastructure and Buildings at the Global Level (2023). Cardona, O.D., Bernal, G.A., Villegas, C.P., Molina, J.F., Herrera, S.A., Marulanda, M.C., Rincón, D.F., Grajales, S., Marulanda, P.M., Gonzalez, D., Maskrey, A. (2023). <https://giri.unepgrid.ch/sites/default/files/2023-11/2.4-INGENIAR-CDRI-Background-Report-Risk-model.pdf>. Accessed 17 Mar 2025.

- MAP, 2025. Malaria Atlas Project. <https://malariaatlas.org/>. Accessed 17 Mar 2025.
- OECD, 2008. Handbook on constructing composite indicators. Methodology and user guide – ISBN 978-92-64-04345. Accessed 17 Mar 2025.
- Pfeffer, D.A., Lucas, T.C., May, D., Harris, J., Rozier, J., Twohig, K.A., Dalrymple, U., Guerra, C.A., Moyes, C.L., Thorn, M., Nguyen, M., et al. 2018. malariaAtlas: an R interface to global malariometric data hosted by the Malaria Atlas Project. *Malaria Journal*, 17(1), p.352
- Reisinger, Andy, Mark Howden, Carolina Vera, et al. (2020) The Concept of Risk in the IPCC Sixth Assessment Report: A Summary of Cross-Working Group Discussions. Intergovernmental Panel on Climate Change, Geneva, Switzerland. pp15. https://www.ipcc.ch/site/assets/uploads/2021/02/Risk-guidance-FINAL_15Feb2021.pdf Accessed 17 Mar 2025
- Láng-Ritter, J., Keskinen, M. & Tenkanen, H. Global gridded population datasets systematically underrepresent rural population. *Nat Commun* **16**, 2170 (2025). <https://doi.org/10.1038/s41467-025-56906-7>. Accessed 26 Mar 2025
- Shen, S. Li, C. van Donkelaar, A. Jacobs, N. Wang, C. Martin, R. V.: Enhancing Global Estimation of Fine Particulate Matter Concentrations by Including Geophysical a Priori Information in Deep Learning. (2024) *ACS ES&T Air*. DOI: 10.1021/acsestair.3c00054 <https://pubs.acs.org/doi/full/10.1021/acsestair.3c00054>. Accessed 17 Mar 2025.
- UNCCD 2024. "Global Sand and Dust Storm Base Map". Technical Brief. https://www.unccd.int/sites/default/files/2024-10/UNCCD-WMO%20Technical%20Brief_SDS%20Source%20Base%20Map_Nov%202024.pdf. Accessed 17 Mar 2025.
- UNICEF 2020. United Nations Children's Fund, A Generation to Protect: Monitoring violence, exploitation and abuse of children within the SDG framework, UNICEF, New York, 2020. Accessed 17 Mar 2025.
- United Nations Office for Disaster Risk Reduction. "Flood hazard and risk assessment." UNDRR, 2020. <https://www.undrr.org/publication/flood-hazard-and-risk-assessment>. Accessed 17 Mar 2025.
- United Nations Office for Disaster Risk Reduction. "Flood." UNDRR, 2020. <https://www.undrr.org/hips-cluster/flood>. Accessed 17 Mar 2025.
- United Nations Office for Disaster Risk Reduction. "Fluvial (Riverine) Flood." UNDRR, 2020. <https://www.undrr.org/understanding-disaster-risk/terminology/hips/mh0007>. Accessed 17 Mar 2025.
- United Nations Office for Disaster Risk Reduction. "Drought." UNDRR, 2020. <https://www.undrr.org/understanding-disaster-risk/terminology/hips/mh0035>. Accessed 17 Mar 2025.
- United Nations Food and Agriculture Organization. "Drought and Agriculture." FAO, 2020. <https://www.fao.org/land-water/water/drought/droughtandag/en/>. Accessed 17 Mar 2025.
- United Nations Environment Programme. "Spreading like Wildfire: The Rising Threat of Extraordinary Landscape Fires." UNEP, 2022. <https://www.unep.org/resources/report/spreading-wildfire-rising-threat-extraordinary-landscape-fires>. Accessed 17 Mar 2025.

- United Nations. "International Day of Combating Sand and Dust Storms." UN, 2023. <https://www.un.org/en/observances/day-of-combating-sand-and-dust-storms>. Accessed 17 Mar 2025.
- United Nations Office for the Coordination of Humanitarian Affairs. "Extreme heat: Preparing for the heatwaves of the future." UN-OCHA, 2022. <https://www.unocha.org/publications/report/world/extreme-heat-preparing-heatwaves-future-october-2022>. Accessed 17 Mar 2025.
- Vousdoukas, M.I., Mentaschi, L., Voukouvalas, E., Verlaan, M., Jevrejeva, S., Jackson, L.P., Feyen, L., 2018. Global probabilistic projections of extreme sea levels show intensification of coastal flood hazard. Nat. Commun. 9, 2360. <https://doi.org/10.1038/s41467-018-04692-w>.
- Vousdoukas, M.I., Mentaschi, L., Voukouvalas, E., Bianchi, A., Dottori, F., Feyen, L. 2018. Climatic and socioeconomic controls of future coastal flood risk in Europe. Nat. Clim. Change. 89 (8), 776–780. <https://doi.org/10.1038/s41558-018-0260-4>.
- World Health Organization. "Malaria." WHO, 2024. <https://www.who.int/en/news-room/fact-sheets/detail/malaria>. Accessed 17 Mar 2025.
- World Health Organization. "Air Pollution." WHO, 2024. <https://www.who.int/news-room/fact-sheets/detail/air-pollution>. Accessed 17 Mar 2025.
- WHO, 2021. World Health Organization global air quality guidelines. Particulate matter, ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide. Geneva: Licence: CCBY-NC-SA 2.0 IGO. <https://www.who.int/publications/i/item/9789240034228>. Accessed 17 Mar 2025.
- World Health Organization. (2024). *Vector-borne diseases*. Retrieved from <https://www.who.int/news-room/fact-sheets/detail/vector-borne-diseases>. Accessed 17 Mar 2025
- Bondarenko M., Priyatikanto R., Zhang W., McKeen T., Cunningham A., Tejedor-Garavito N., Woods T., Hilton J., Cihan D., Nosatiuk B., Brinkhoff T., Tatem A., Sorichetta A. Unconstrained and constrained estimates of 2024 total number of people under the age of 18 per grid square at a resolution of 3 arc (approximately 100m at the equator), R2024A version v1.0 WorldPop, University of Southampton DOI: 10.5258/SOTON/WP00799
- World Meteorological Organization. "Tropical Storm Classification." <https://wmo.int/content/classification-of-tropical-cyclones>. Accessed 17 Mar 2025.
- WMO, 2011. Manual on Flood Forecasting and Warning. WMO-No. 1072. World Meteorological Organization (WMO). Accessed 17 Mar 2025.
- WMO, 2012. Definition number 543. International Glossary of Hydrology. WMO-No. 385. World Meteorological Organization (WMO). Accessed 17 Mar 2025.
- WMO, 2025. Fires. Global Climate Observing System. <https://gcos.wmo.int/site/global-climate-observing-system-gcos/essential-climate-variables/fire>. Accessed 17 Mar 2025.
- World malaria report 2024: addressing inequity in the global malaria response. Geneva: World Health Organization; 2024. License: CC BY-NC-SA 3.0 IGO.
- Wilks, D. S. (2011). *Statistical Methods in the Atmospheric Sciences* (3rd ed.). Academic Press.
- World Meteorological Organization (WMO). (2009). *Guidelines on Analysis of Extremes in a Changing Climate in Support of Adaptation Decision-Making* (WMO-TD No. 1500). Geneva, Switzerland.